



SUSTAINABLE OLYMPIAD

Sustainable Olympiad 2024

Round 1: Online Knowledge Challenge

Senior Division (ages 15–18)

Time allowed	90 minutes
Total marks	60
Part A	10 multiple-choice questions, 20 marks
Part B	5 structured problems, 40 marks

Instructions

- Answer **all** questions. Work together as a team.
- Part A: choose **one** answer (A, B, C or D) for each question. There are no negative marks.
- Part B: show all your working. Marks are given for correct method even if the final answer is wrong.
- Give numerical answers to an appropriate number of significant figures, with units.
- Scientific calculators are allowed. Internet access and AI tools are not allowed during the round.
- A data sheet is provided on the next page.

Team name

School

Data sheet

- Acceleration due to gravity: $g = 9.81 \text{ m s}^{-2}$
- Specific heat capacity of water: $c = 4.18 \text{ kJ kg}^{-1} \text{ K}^{-1}$; density of water 1.00 kg L^{-1}
- $1 \text{ kWh} = 3.6 \text{ MJ}$; $0 \text{ }^{\circ}\text{C} = 273.15 \text{ K}$; $1 \text{ year} = 8760 \text{ h}$
- Planck constant $h = 6.63 \times 10^{-34} \text{ J s}$; speed of light $c = 3.00 \times 10^8 \text{ m s}^{-1}$; $1 \text{ eV} = 1.60 \times 10^{-19} \text{ J}$
- Stefan-Boltzmann constant $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
- Molar masses (g mol^{-1}): H 1.008, C 12.01, N 14.01, O 16.00
- Solar constant $S = 1361 \text{ W m}^{-2}$; GWP_{100} of $\text{N}_2\text{O} = 273$

Part A: Multiple choice (20 marks)

Each question is worth 2 marks. Choose one answer.

A1. A home battery stores 13.5 kWh and has a round-trip efficiency of 90%. [2]
How much energy must be supplied while charging so that it can deliver 13.5 kWh?

- A** 12.2 kWh
- B** 13.5 kWh
- C** 15.0 kWh
- D** 24.3 kWh

A2. The band gap of crystalline silicon is 1.12 eV. What is the longest [2]
wavelength of light that can create an electron-hole pair in silicon?

- A** 443 nm
- B** 700 nm
- C** 1110 nm
- D** 1550 nm

A3. What is the approximate maximum efficiency of a single-junction solar cell [2]
under ordinary sunlight (the Shockley-Queisser limit)?

- A** 15%
- B** 33%
- C** 59%
- D** 86%

A4. Which organisms fix atmospheric nitrogen in the root nodules of legumes [2]
such as beans and clover?

- A** Nitrifying bacteria such as *Nitrosomonas*
- B** Denitrifying bacteria such as *Pseudomonas*
- C** *Rhizobium* bacteria
- D** Mycorrhizal fungi

- A5.** A river sample has a high biochemical oxygen demand (BOD). This most likely indicates: [2]
- A** a high concentration of dissolved oxygen
 - B** a large amount of biodegradable organic matter, for example from sewage
 - C** a low concentration of nitrate
 - D** a high salinity
- A6.** A pumped-storage reservoir holds $1.0 \times 10^6 \text{ m}^3$ of water at an average height of 300 m above the turbines. How much gravitational potential energy is stored? [2]
- A** 8.2 MWh
 - B** 82 MWh
 - C** 818 MWh
 - D** 8.2 GWh
- A7.** Global demand for a resource grows at 7% per year. Approximately how long does it take for demand to double? [2]
- A** 7 years
 - B** 10 years
 - C** 14 years
 - D** 70 years
- A8.** Taking Earth's albedo as 0.30 and ignoring the atmosphere, what is Earth's effective radiating temperature? [2]
- A** About -18°C
 - B** About 0°C
 - C** About $+15^\circ\text{C}$
 - D** About $+33^\circ\text{C}$
- A9.** How much energy is needed to heat 150 L of water from 15°C to 55°C ? [2]
- A** 0.70 kWh
 - B** 6.97 kWh
 - C** 25.1 kWh
 - D** 69.7 kWh
- A10.** What is meant by 'green hydrogen'? [2]
- A** Hydrogen made from natural gas, with the CO_2 captured and stored
 - B** Hydrogen made by electrolysis of water using renewable electricity
 - C** Hydrogen made by gasifying coal
 - D** Hydrogen made by steam reforming of methane without carbon capture

Part B: Structured problems (40 marks)

B1. Reusable or single-use? (9 marks)

Manufacturing a reusable glass bottle causes 0.40 kg CO₂e. Each use cycle (collection, washing and refilling) adds 0.03 kg CO₂e. A single-use aluminium can causes 0.17 kg CO₂e per use, including recycling.

- (a) Find the minimum number of uses for the bottle's footprint per use to be lower than the can's. [3]

- (b) Calculate the footprint per use if a bottle is used 20 times, and the percentage saving compared with the can. [4]

- (c) In reality, after each use a bottle is returned with probability $p = 0.90$, otherwise it is lost. Show that the expected number of uses is $1/(1 - p)$, and calculate the expected footprint per use. [4]

B2. A solar farm (8 marks)

A solar farm has a peak capacity of 50 MW and a capacity factor of 18%. An average household uses 3500 kWh of electricity per year. Electricity from coal emits 0.95 kg CO₂ per kWh.

- (a) Calculate the farm's annual electricity output, in GWh. [2]

(b) How many average households could this supply? **[2]**

(c) How much CO_2 per year is avoided if this output replaces coal-fired electricity? **[2]**

(d) Give two reasons why the capacity factor of a solar farm is far below 100%. **[2]**

B3. Sewage effluent in a river (8 marks)

Upstream of a treatment plant, a river flows at $4.0 \text{ m}^3 \text{ s}^{-1}$ with a BOD of 2.0 mg L^{-1} and dissolved oxygen (DO) of 9.0 mg L^{-1} . The plant discharges $0.50 \text{ m}^3 \text{ s}^{-1}$ of effluent with a BOD of 60 mg L^{-1} and DO of 2.0 mg L^{-1} . Assume complete mixing.

(a) Calculate the BOD of the river just below the outlet. **[2]**

(b) Calculate the dissolved oxygen just below the outlet. **[2]**

(c) Ignoring re-aeration, would the DO stay above 5.0 mg L^{-1} (needed by most fish) once all the BOD has been exerted? Explain. **[2]**

(d) What is the maximum effluent BOD that would keep the DO at or above 5.0 mg L⁻¹? **[2]**

B4. Fertilisers and nitrous oxide (8 marks)

Ammonium nitrate (NH₄NO₃) and urea (CO(NH₂)₂) are common nitrogen fertilisers.

(a) Calculate the percentage by mass of nitrogen in each fertiliser. **[2]**

(b) A farmer applies 120 kg of nitrogen per hectare to 15 ha using urea. What mass of urea is needed? **[2]**

(c) 1.0% of the applied nitrogen is emitted as nitrogen in N₂O. Calculate the mass of N₂O and its CO₂-equivalent. **[3]**

(d) Suggest one way to reduce these emissions without reducing crop yield. **[1]**

B5. How long will it last? (7 marks)

Known reserves of a metal are 880 Mt, and current annual consumption is 26 Mt.

(a) Calculate the static lifetime of the reserves. **[1]**

(b) If consumption grows continuously at 3.0% per year, cumulative use after t years is $(C_0/r)(e^{rt} - 1)$. Calculate when the reserves run out. **[4]**

(c) Give two reasons why the real lifetime could be longer than either estimate. **[2]**

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